the continental edge. Alternatively, a trench may be formed between two plates.

In some cases when two continental plates converge, neither plate can be forced under the other. Instead, folds may be created. Great mountain ranges like the Himalayas have been formed in this way.

The movement of these plates causes changes on the surface of the earth. The earth movements are divided on the basis of the forces which cause them. The forces which act in the interior of the earth are called as Endogenic forces and the forces that work on the surface of the earth are called as Exogenic forces.

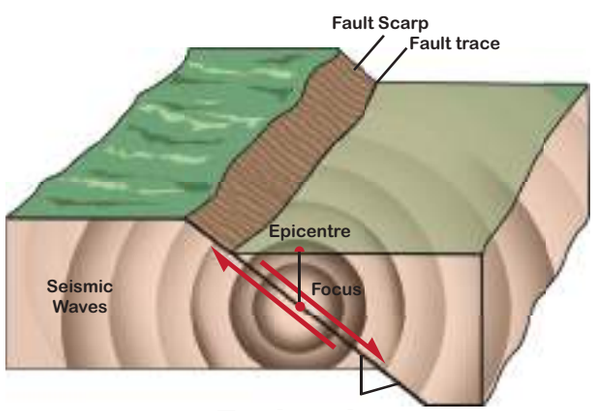
Endogenic forces produce sudden movements and Exogenic forces produce slow movements. Endogenic movements produce earthquakes and volcanoes that cause mass destruction over the surface of the earth.

DO YOU KNOW?

The asthenosphere is the part of the mantle that flows and moves the plates of the earth.

Earthquake

A sudden movement of a portion of the earth's crust which produces a shaking or trembling is known as an earthquake. The point where these vibrations originate is called the focus of the earthquake. The point of the earth's surface directly above the focus is called the epicentre of the earthquake. From the focus, the earthquake vibrations travel in different directions in the form of seismic waves.



Earthquake

The earthquake waves are recorded by an instrument known as seismograph. The magnitude of an earthquake is measured by the Richter scale. The units on this scale range from 0 to 9.

Causes of Earthquake

The chief cause of earthquake is the sudden slipping of the portion of the earth's crust along fractures or faults. The movement of the molten rocks underneath the surface produce strains which break the rocks apart. The sudden shifting of landmass causes upheavals in the crust of the earth sending vibrations or waves into the surrounding portions of the earth. Sometimes the surface of the earth itself cracks.

Effects of Earthquakes

Earthquakes may cause changes in the earth's surface. Vibrations often set landslides in mountainous regions. A greater danger in an earthquake is the falling of buildings. Most of the houses which collapsed were made of mud and bricks and proved to be death traps. Underground water system is naturally disturbed by such movements. Fire is another great danger.

There are three types of earthquake waves:
P waves or longitudinal waves
S waves or transverse waves
L waves or surface waves

An earthquake which originates below or near the sea causes great disturbance in the water. The floods and waves cause great loss of life, sometimes more than the earthquake

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